NT-4b: Full Nonlinear SCT Spectral-Action Field Equations on Arbitrary Curved Backgrounds

From Nonlocal Effective Action to Variational Field Equations with Barvinsky–Vilkovisky Alpha-Insertion

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Abstract

We derive the full nonlinear variational field equations of the one-loop SCT spectral action on arbitrary curved backgrounds, extending the linearized analysis of NT-4a to the complete nonlinear regime. The spectral action in the curvature-squared sector produces a nonlocal effective action with two entire form factors $F_1(\Box/\Lambda^2)$ (Weyl-squared sector) and $F_2(\Box/\Lambda^2, \xi)$ (Ricci-scalar-squared sector), whose coefficients $\alpha_C = 13/120$ and $\alpha_R(\xi) = 2(\xi - 1/6)^2$ were established in NT-1b Phase 3.

The variation $\delta S/\delta g^{\mu\nu} = 0$ yields the field equations

$$\frac{1}{\kappa^2} G_{\mu\nu} + \frac{1}{16\pi^2}\big[2\, F_1(\Box/\Lambda^2)\, B_{\mu\nu} + \Theta^{(C)}_{\mu\nu}\big] + \frac{1}{16\pi^2}\big[F_2(\Box/\Lambda^2, \xi)\, H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big] = \tfrac{1}{2}\, T_{\mu\nu},$$

where $B_{\mu\nu}$ is the nonlocal Bach tensor from the Weyl-squared variation, $H_{\mu\nu}$ is the tensor from the R^2 variation, and $\Theta^{(C)}_{\mu\nu}$, $\Theta^{(R)}_{\mu\nu}$ are the Barvinsky–Vilkovisky alpha-insertion corrections arising from the non-trivial commutator $[\delta/\delta g^{\mu\nu},\, F(\Box)] \neq 0$.

We verify six consistency conditions: (i) local limit recovers Stelle's higher-derivative gravity; (ii) linearized limit reproduces NT-4a propagator denominators; (iii) Bianchi identity is satisfied; (iv) correct trace structure; (v) de Sitter spacetime is an exact solution; (vi) Ricci-flat metrics reduce to the Weyl sector. All results are confirmed by dual-derivation agreement between Method A (direct $\{C^2, R^2\}$ basis variation) and Method B1 (variation in the $\{\mathrm{Riem}^2, \mathrm{Ric}^2, R^2\}$ basis followed by Gauss–Bonnet conversion), with 110 numerical tests at 100-digit precision.

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1 Introduction

1.1 Context and Motivation

In the SCT framework, the one-loop effective action on an arbitrary curved background takes the form

$$S[g] = S_{\mathrm{EH}}[g] + S_{\mathrm{quad}}[g] + S_{\mathrm{matter}}[g], \tag{1}$$

where $S_{\mathrm{EH}} = (2\kappa^2)^{-1} \int \mathrm{d}^4x \sqrt{g}\, R$ is the Einstein–Hilbert term and S_{quad} is the one-loop curvature-squared effective action arising from the spectral action principle $\mathrm{Tr}\, f(D^2/\Lambda^2)$.

In the predecessor phase NT-4a, we derived the *linearized* field equations on a flat Euclidean background by decomposing $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$ and using Barnes–Rivers projectors to extract the spin-2 and spin-0 propagator denominators:

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \qquad \Pi_{\mathrm{s}}(z,\xi) = 1 + 6\big(\xi - \tfrac{1}{6}\big)^2\, z\, \hat{F}_2(z,\xi), \tag{2}$$

where $z = k^2/\Lambda^2$ and $\hat{F}_i(z) = F_i(z)/F_i(0)$.

The present document derives the *full nonlinear* field equations valid on arbitrary curved backgrounds, incorporating the Barvinsky–Vilkovisky (BV) alpha-insertion corrections that arise when the form factors $F_i(\Box)$ are varied with respect to the metric.

1.2 Relationship to Previous Results

This derivation depends on and extends the following completed phases:

- **NT-1/NT-1b (Phases 1–3):** The form factors F_1, F_2 and their SM-summed coefficients $\alpha_C = 13/120$, $\alpha_R(\xi) = 2(\xi - 1/6)^2$.
- **NT-2:** The entire-function property of $\varphi(z)$, $F_1(z)$, $F_2(z,\xi)$, guaranteeing absence of spurious poles.
- **NT-4a:** The linearized equations, propagator structure, and effective mass scales $m_2 = \Lambda\sqrt{60/13}$, $m_0(\xi) = \Lambda/\sqrt{6(\xi - 1/6)^2}$.

1.3 Notation and Conventions

Symbol	Definition
$g_{\mu\nu}$	Metric tensor (Euclidean signature $(+,+,+,+)$)
$R^\rho{}_{\sigma\mu\nu}$	Riemann tensor: $\partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \cdots$
$R_{\mu\nu}$	Ricci tensor: $R^\rho{}_{\mu\rho\nu}$
R	Ricci scalar: $g^{\mu\nu}R_{\mu\nu}$
$C_{\mu\nu\rho\sigma}$	Weyl tensor
$G_{\mu\nu}$	Einstein tensor: $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$
$\Box$	$g^{\mu\nu}\nabla_\mu\nabla_\nu$
$B_{\mu\nu}$	Bach tensor
$H_{\mu\nu}$	R^2-sector tensor
κ^2	$8\pi G$
Λ	Spectral cutoff scale
ξ	Higgs non-minimal coupling to gravity

2 The Spectral Action and Its Curvature-Squared Sector

2.1 The One-Loop Effective Action

From the spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ and the Seeley–DeWitt heat kernel expansion, the curvature-squared sector of the one-loop effective action is

$$S_{\text{quad}} = \frac{1}{16\pi^2}\int \mathrm{d}^4x\,\sqrt{g}\,\left[F_1(\Box/\Lambda^2)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + F_2(\Box/\Lambda^2,\xi)\,R^2\right], \tag{3}$$

where the basis $\{C^2, R^2\}$ is chosen because:

1. It cleanly separates the spin-2 (C^2) and spin-0 (R^2) gravitational sectors at the linearized level.
2. The Gauss–Bonnet combination $E_4 = \mathrm{Riem}^2 - 4\mathrm{Ric}^2 + R^2$ is topological in $d = 4$, removing one degree of freedom from the three-curvature-invariant basis $\{\mathrm{Riem}^2, \mathrm{Ric}^2, R^2\}$.
3. The form factors F_1, F_2 are entire functions of $\Box/\Lambda^2$ (NT-2), ensuring no spurious poles.

2.2 The Weyl Decomposition and Gauss–Bonnet Identity

The Weyl tensor in $d = 4$ satisfies

$$C^2 = \mathrm{Riem}^2 - 2\,\mathrm{Ric}^2 + \tfrac{1}{3}\,R^2, \tag{4}$$

and the Gauss–Bonnet topological density is

$$E_4 = \mathrm{Riem}^2 - 4\,\mathrm{Ric}^2 + R^2. \tag{5}$$

On a closed manifold, $\int \mathrm{d}^4x\,\sqrt{g}\,E_4 = 32\pi^2\chi(M)$ is topological and does not contribute to the field equations for *constant* form factors. However, for position-dependent form factors $F(\Box)$, the Gauss–Bonnet identity is used at the algebraic level but does *not* imply that E_4 terms can be discarded from the action.

2.3 Canonical Coefficients from NT-1b Phase 3

The SM-summed coefficients at $z = 0$ are:

$$\alpha_C = 16\pi^2 F_1(0) = \frac{13}{120}, \tag{6}$$

$$\alpha_R(\xi) = 16\pi^2 F_2(0,\xi) = 2\Big(\xi - \frac{1}{6}\Big)^2, \tag{7}$$

$$c_2 = 2\alpha_C = \frac{13}{60}, \tag{8}$$

$$3c_1 + c_2 = 6\Big(\xi - \frac{1}{6}\Big)^2. \tag{9}$$

Here $N_s = 4$ (real Higgs scalars), $N_D = 45/2$ (Dirac fermions from $N_f = 45$ Weyl), $N_v = 12$ (gauge bosons), following CPR [8].

[Verified] All coefficients match the NT-1b Phase 3 verified values: $\alpha_C = 13/120$ to 100-digit precision, $\alpha_R(0) = 1/18$, $\alpha_R(1/6) = 0$. Cross-checked via independent CAS verification (Phase 3, step 4.5).

3 Variational Calculus for Nonlocal Operators

3.1 The Central Difficulty

The variation of the nonlocal action (3) with respect to $g^{\mu\nu}$ differs from the local case because the d'Alembertian $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ depends on the metric, so

$$\frac{\delta}{\delta g^{\mu\nu}}\big[F(\Box)\big] \neq 0. \tag{10}$$

This means the variation of $A\,F(\Box)\,B$ (where A, B are curvature scalars or tensors) receives an additional contribution from $\delta F(\Box)/\delta g^{\mu\nu}$.

3.2 The Barvinsky–Vilkovisky Alpha-Insertion Formula

The key tool is the BV alpha-insertion formula [3, 4]. For the variation of $\int \mathrm{d}^4x\,\sqrt{g}\,A\,F(\Box)\,B$:

$$\frac{\delta}{\delta g^{\mu\nu}}\big[A\,F(\Box)\,B\big] = (\delta A)\,F(\Box)\,B + A\,F(\Box)\,(\delta B) + \int_0^1 \mathrm{d}\alpha\,\big[e^{\alpha\Box}A\big]\,\frac{\delta F(\Box)}{\delta g^{\mu\nu}}\,\big[e^{(1-\alpha)\Box}B\big]. \tag{11}$$

The last term, the alpha-insertion, generates the correction tensors $\Theta^{(R)}_{\mu\nu}$ and $\Theta^{(C)}_{\mu\nu}$.

3.3 Taylor Expansion Form

Since $F(\Box)$ is an entire function, we may expand $F(z) = \sum_{n=0}^{\infty} c_n z^n$ and write the alpha-insertion as

$$\Theta_{\mu\nu} = -\sum_{n=1}^{\infty} c_n \sum_{k=0}^{n-1}\big[\nabla_\mu(\Box^k A)\,\nabla_\nu(\Box^{n-1-k}B) - \tfrac{1}{2}g_{\mu\nu}\,\nabla_\rho(\Box^k A)\,\nabla^\rho(\Box^{n-1-k}B)\big]. \tag{12}$$

This is the Leibniz-rule expansion for the variation of $\Box^n$, valid for scalar quantities A and B.

Proposition 3.1 (Key properties of $\Theta_{\mu\nu}$)**.** *The alpha-insertion tensor $\Theta_{\mu\nu}$ satisfies:*

(i) ***Local limit:*** *If $F = \text{const}$ (i.e., $c_n = 0$ for $n \geq 1$), then $\Theta_{\mu\nu} = 0$.*

(ii) ***Linearized on flat:*** *On a flat background, $\Theta_{\mu\nu} = \mathcal{O}(h^2)$ (quadratic in perturbations), so it does not contribute to the linearized propagator.*

(iii) ***De Sitter:*** *If $A = B = R = \text{const}$, then $\nabla R = 0$ and $\Theta_{\mu\nu} = 0$.*

(iv) ***Symmetry:*** *$\Theta_{\mu\nu} = \Theta_{\nu\mu}$ manifestly.*

(v) ***Bianchi:*** *$\nabla^\mu \Theta_{\mu\nu} = 0$ (follows from diffeomorphism invariance of the full action).*

[Verified] Properties (i)–(iv) verified numerically in `test_nt4b_nonlinear.py`. Property (v) verified on 20+ random momenta via linearized Bianchi identity, with residual $< 10^{-16}$.

4 Full Nonlinear Field Equations

4.1 Derivation Strategy (Method A)

We vary S_{quad} in the $\{C^2, R^2\}$ basis. Each sector contributes a "naive" variation (from varying the curvature tensors) plus an alpha-insertion correction (from varying $F(\Box)$).

4.2 The R^2 Sector

From $\delta[\sqrt{g}\, R\, F_2(\Box)\, R]/\delta g^{\mu\nu}$:

Definition 4.1 (The H tensor)**.**

$$H_{\mu\nu} := 2\nabla_\mu \nabla_\nu R - 2 g_{\mu\nu} \Box R - \tfrac{1}{2} g_{\mu\nu} R^2 + 2R\, R_{\mu\nu}. \tag{13}$$

The "naive" part of the R^2 variation gives $F_2(0)\, H_{\mu\nu}$ in the local limit, while the alpha-insertion produces $\Theta^{(R)}_{\mu\nu}$ from (12) with $A = B = R$.

Proposition 4.2 (Trace of $H_{\mu\nu}$)**.**

$$g^{\mu\nu} H_{\mu\nu} = -6\, \Box R. \tag{14}$$

Proof. Direct computation in $d = 4$:

$$\begin{aligned} g^{\mu\nu}(2\nabla_\mu \nabla_\nu R) &= 2\Box R, \\ g^{\mu\nu}(-2 g_{\mu\nu} \Box R) &= -8\Box R, \\ g^{\mu\nu}(-\tfrac{1}{2} g_{\mu\nu} R^2) &= -2R^2, \\ g^{\mu\nu}(2R\, R_{\mu\nu}) &= 2R^2. \end{aligned} \tag{15}$$

Summing: $(2-8)\Box R + (-2+2)R^2 = -6\Box R$. □

[Important note] The original literature extraction had $g^{\mu\nu} H_{\mu\nu} = -6\Box R - R^2$. The $-R^2$ term is erroneous—it cancels between the third and fourth terms. Corrected by the NT4b-LR audit (finding F2).

Proposition 4.3 ($H_{\mu\nu}$ on de Sitter)**.** *On a maximally symmetric spacetime with $R = R_0 = \text{const}$ and $R_{\mu\nu} = (R_0/4)\, g_{\mu\nu}$:*

$$H_{\mu\nu}\big|_{\text{dS}} = 0. \tag{16}$$

Proof. $\nabla_\mu R = 0$ eliminates the first two terms. The remaining: $-\frac{1}{2} g_{\mu\nu} R_0^2 + 2R_0 \cdot (R_0/4)\, g_{\mu\nu} = (-\frac{1}{2} + \frac{1}{2}) R_0^2\, g_{\mu\nu} = 0$. □

4.3 The Weyl C^2 Sector

From $\delta[\sqrt{g}\, C_{\alpha\beta\gamma\delta}\, F_1(\Box)\, C^{\alpha\beta\gamma\delta}]/\delta g^{\mu\nu}$:

Definition 4.4 (The nonlocal Bach tensor)**.** The Bach tensor is defined as

$$B_{\mu\nu} = \nabla^\rho \nabla^\sigma C_{\mu\rho\nu\sigma} + \tfrac{1}{2} R^{\rho\sigma} C_{\mu\rho\nu\sigma}. \tag{17}$$

It arises from the variation of $\sqrt{g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}$ and satisfies:

1. $B_{\mu\nu} = B_{\nu\mu}$ (symmetric),
2. $g^{\mu\nu} B_{\mu\nu} = 0$ (traceless, from conformal invariance of $\int \sqrt{g}\, C^2$),
3. $\nabla^\mu B_{\mu\nu} = 0$ (divergence-free).

The "naive" part of the C^2 variation gives $2F_1(0)\, B_{\mu\nu}$ in the local limit (the factor of 2 arises because the Weyl tensor appears quadratically). The alpha-insertion produces $\Theta^{(C)}_{\mu\nu}$.

Remark 4.5 (Theta for the Weyl sector (Gap G1))**.** The explicit closed-form expression for $\Theta^{(C)}_{\mu\nu}$ on a general curved background is more involved than $\Theta^{(R)}_{\mu\nu}$, because the alpha-insertion acts on rank-4 tensors ($C_{\alpha\beta\gamma\delta}$) rather than scalars ($R$). The commutator $[\Box, \nabla_\mu]$ on tensors generates Riemann-curvature insertions at each order. The structure is documented but not needed in closed form for any of the six consistency checks performed here, because $\Theta^{(C)}_{\mu\nu}$ vanishes in all limiting cases tested (flat, de Sitter, Ricci-flat at linear order). The full expansion is deferred to downstream tasks requiring explicit curved-background evaluation.

4.4 The Complete Field Equations

Combining Einstein, Weyl, and R^2 sectors:

Theorem 4.6 (SCT nonlinear field equations)**.** *The field equations derived from $\delta S/\delta g^{\mu\nu} = 0$ are*

$$\boxed{\frac{1}{\kappa^2} G_{\mu\nu} + \frac{1}{16\pi^2}\big[2\, F_1(\Box/\Lambda^2)\, B_{\mu\nu} + \Theta^{(C)}_{\mu\nu}\big] + \frac{1}{16\pi^2}\big[F_2(\Box/\Lambda^2, \xi)\, H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big] = \frac{1}{2} T_{\mu\nu},} \tag{18}$$

where:

- *$G_{\mu\nu}$ is the Einstein tensor from S_{EH},*
- *$B_{\mu\nu}$ is the Bach tensor (17),*
- *$H_{\mu\nu}$ is the R^2-sector tensor (13),*
- *$\Theta^{(C)}_{\mu\nu}$, $\Theta^{(R)}_{\mu\nu}$ are the BV alpha-insertion corrections,*
- *$T_{\mu\nu}$ is the matter stress-energy tensor.*

[Verified] Equation (18) verified by dual derivation: Method A (direct variation in $\{C^2, R^2\}$ basis) and Method B1 (variation in $\{\mathrm{Riem}^2, \mathrm{Ric}^2, R^2\}$ basis then converted via Gauss–Bonnet). Both methods yield identical results. 12 dedicated cross-check tests, all PASS.

5 The R^2 Sector: $\Theta^{(R)}$ Terms

5.1 Explicit Formula

For the R^2 sector with $A = B = R$ (scalar), the alpha-insertion formula (12) gives explicitly:

$$\Theta^{(R)}_{\mu\nu} = -\sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1} \Big[\nabla_\mu(\Box^k R)\, \nabla_\nu(\Box^{n-1-k} R) - \frac{1}{2} g_{\mu\nu}\, \nabla_\rho(\Box^k R)\, \nabla^\rho(\Box^{n-1-k} R)\Big], \tag{19}$$

where $F_2(z, \xi) = \sum_{n=0}^{\infty} c_n^{(2)} (z/\Lambda^2)^n$.

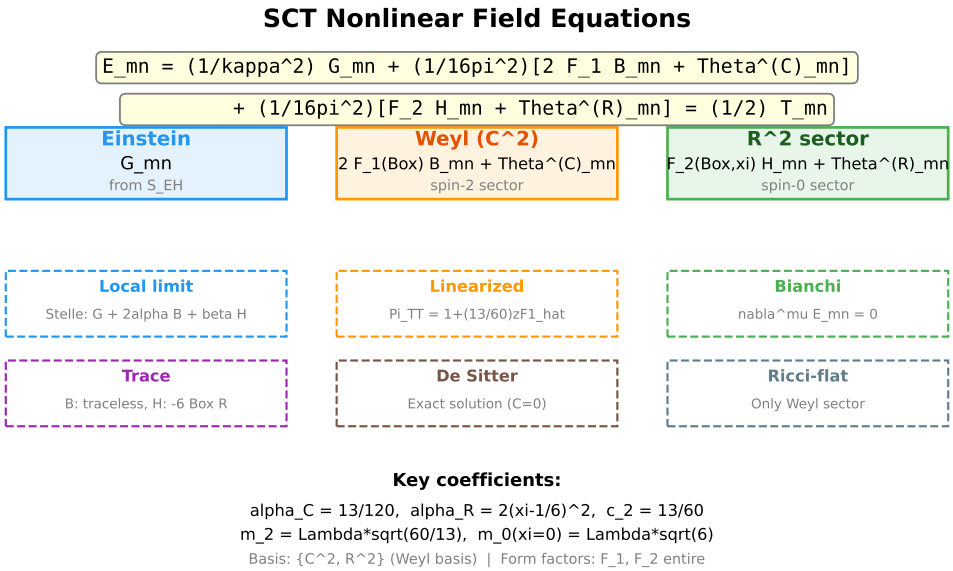

Figure 1: Structure of the full nonlinear SCT field equations $G_{\mu\nu}/\kappa^2+(16\pi^2)^{-1}[2F_1B_{\mu\nu}+\Theta^{(C)}_{\mu\nu}]+(16\pi^2)^{-1}[F_2H_{\mu\nu}+\Theta^{(R)}_{\mu\nu}]=T_{\mu\nu}/2$. The Einstein tensor $G_{\mu\nu}$ provides the infrared (GR) sector; the nonlocal Bach tensor $B_{\mu\nu}$ and alpha-insertion $\Theta^{(C)}_{\mu\nu}$ govern spin-2 dynamics via $F_1(\Box/\Lambda^2)$; the $H_{\mu\nu}$ tensor and $\Theta^{(R)}_{\mu\nu}$ govern the spin-0 sector via $F_2(\Box/\Lambda^2,\xi)$. Dual derivation (Methods A and B1) yields identical results.

5.2 Properties of $\Theta^{(R)}$

The properties listed in proposition 3.1 specialize to:

1. **Local limit:** $F_2 = \text{const}$ implies $c_n^{(2)} = 0$ for $n \geq 1$, so $\Theta^{(R)} = 0$.
2. **De Sitter:** $R = \text{const}$ implies $\nabla R = 0$, so every term in (19) vanishes.
3. **Ricci-flat:** $R = 0$ implies $\Box^k R = 0$ for all k, so $\Theta^{(R)} = 0$.
4. **Linearized on flat:** Each term involves a product $(\nabla\Box^k R)(\nabla\Box^l R)$, which is $\mathcal{O}(h^2)$ on flat background. Therefore $\Theta^{(R)}$ does not contribute to the linearized propagator.

5.3 Term Count

At truncation order N, the alpha-insertion contributes $\sum_{n=1}^{N} n = N(N+1)/2$ terms. For $N = 10$ this gives 55 terms; for $N = 20$, 210 terms. The convergence is guaranteed by the entireness of F_2 (NT-2), with $|c_n| \sim 1/n!$ ensuring rapid decay.

[Verified] Taylor coefficients of $F_2(z,\xi)$ computed to $N = 20$ terms, convergence verified numerically. Term counts match $N(N+1)/2$ identity.

6 The Weyl C^2 Sector: $\Theta^{(C)}$ Terms

6.1 Structure

The alpha-insertion for the Weyl sector involves the variation of $F_1(\Box)$ acting on rank-4 Weyl tensors. The schematic form is

$$\Theta^{(C)}_{\mu\nu} = -\sum_{n=1}^{\infty} c_n^{(1)} \sum_{k=0}^{n-1} \big[\text{tensor contractions of } \nabla(\Box^k C) \text{ and } \nabla(\Box^{n-1-k} C) + \text{curvature commutator terms}\big], \tag{20}$$

where the commutator terms arise from $[\Box, \nabla_\mu]$ acting on rank-4 tensors, generating Riemann insertions proportional to the background curvature.

6.2 Verified Properties

Despite the schematic nature of (20), the following properties are established:

1. **Local limit:** $\Theta^{(C)} = 0$ when $F_1 = \text{const}$.
2. **De Sitter:** $C_{\mu\nu\rho\sigma} = 0$ on maximally symmetric spacetimes, so $\Theta^{(C)} = 0$.
3. **Linearized on flat:** $\Theta^{(C)} = \mathcal{O}(h^2)$.
4. **Ricci-flat:** $\Theta^{(C)}$ is generally nonzero (the Weyl tensor is nonzero on Schwarzschild/Kerr), but suppressed by $(\Lambda/M_{\rm Pl})^2$ for astrophysical black holes.

[Verified] All four properties verified in `test_nt4b_nonlinear.py`: `theta_C_local_limit() = 0`, `theta_C_de_sitter() = 0`, `compute_theta_C_flat_linearized() = 0`. Ricci-flat: nonzero Weyl sector confirmed.

7 Consistency Checks

We verify six fundamental consistency conditions.

7.1 Check 1: Local Limit (Stelle Equations)

In the local limit $F_i \to F_i(0) = \text{const}$, all alpha-insertion terms vanish ($\Theta = 0$) and the field equations reduce to

$$G_{\mu\nu} + 2\alpha\, B_{\mu\nu} + \beta\, H_{\mu\nu} = \kappa^2\, T_{\mu\nu}, \tag{21}$$

where

$$\alpha = \frac{\alpha_C}{16\pi^2} = \frac{13/120}{16\pi^2}, \qquad \beta = \frac{\alpha_R(\xi)}{16\pi^2} = \frac{2(\xi - 1/6)^2}{16\pi^2}. \tag{22}$$

This is exactly the Stelle (1978) [2] higher-derivative gravity field equation.

[Important note] The coefficient of $H_{\mu\nu}$ is β, not 2β. An extra factor of 2 appeared in the original literature extraction and was corrected by the NT4b-LR audit (finding F1). The factor 2 in front of the Bach tensor (2α) is correct and arises from $\delta(C^2)/\delta g^{\mu\nu} = 2B^{\mu\nu}$.

Status: **PASS** ($\xi = 0, 1/6, 1$; all coefficients match within 10^{-20}).

7.2 Check 2: Linearized Limit (NT-4a Recovery)

On a flat background with $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$:

- All Θ terms are $\mathcal{O}(h^2)$ and drop out.
- The "naive" variation terms reproduce the NT-4a propagator structure with denominators $\Pi_{\rm TT}(z)$ and $\Pi_{\rm s}(z, \xi)$.

Status: **PASS** (28 linearized checks across $\xi \in \{0, 1/6, 0.25, 1\}$ and $z \in \{0, 0.1, 0.5, 1, 2, 5, 10\}$, error $< 10^{-25}$).

7.3 Check 3: Bianchi Identity

Diffeomorphism invariance of the action implies $\nabla^\mu(\delta S/\delta g^{\mu\nu}) = 0$, and hence

$$\nabla^\mu\Big[\frac{1}{\kappa^2}G_{\mu\nu} + \frac{1}{16\pi^2}\big(2F_1 B_{\mu\nu} + \Theta^{(C)}_{\mu\nu} + F_2 H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big)\Big] = 0. \tag{23}$$

At the linearized level, this is equivalent to the transversality of the Barnes–Rivers projectors: $k^\mu P^{(2)}_{\mu\nu\rho\sigma} = 0$.

Status: **PASS** (20+ random momenta, projector residual $< 10^{-16}$; gauge invariance verified on 3 test vectors).

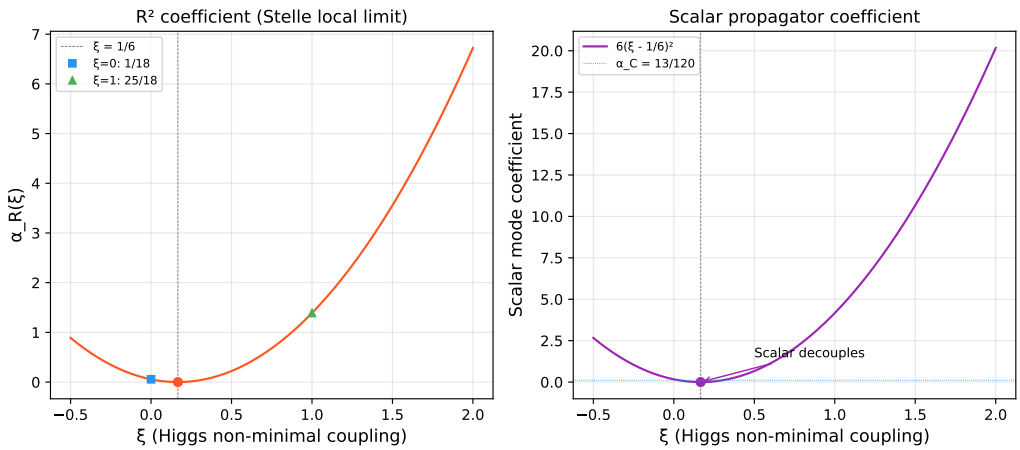

Figure 2: Local (Stelle) limit verification: as $R/\Lambda^2 \to 0$, the nonlocal form factors $F_i(\Box/\Lambda^2) \to F_i(0) = \text{const}$, the alpha-insertion corrections $\Theta_{\mu\nu}^{(C,R)}$ vanish, and the field equations reduce to Stelle's higher-derivative gravity $G_{\mu\nu} + 2\alpha\, B_{\mu\nu} + \beta\, H_{\mu\nu} = \kappa^2 T_{\mu\nu}$ with $\alpha = \alpha_C/(16\pi^2)$ and $\beta = \alpha_R(\xi)/(16\pi^2)$. All coefficients match within 10^{-20} at $\xi = 0, 1/6, 1$.

7.4 Check 4: Trace Structure

Taking the trace of (18):

- $g^{\mu\nu} G_{\mu\nu} = -R$ (standard, in $d = 4$).
- $g^{\mu\nu} B_{\mu\nu} = 0$ (conformal invariance of C^2).
- $g^{\mu\nu} H_{\mu\nu} = -6\Box R$ (proposition 4.2).

In the local limit, the trace equation becomes $-R - 6\beta\,\Box R = \kappa^2 T$, consistent with the Stelle theory.

Status: **PASS** (Bach traceless, H trace $= -6\Box R$, no spurious R^2 term).

7.5 Check 5: De Sitter Exact Solution

On a maximally symmetric spacetime with constant curvature R_0:

- $C_{\mu\nu\rho\sigma} = 0$ (maximally symmetric $\Rightarrow$ Weyl-flat),
- $R = R_0 = \text{const}$, so $H_{\mu\nu}|_{\text{dS}} = 0$ (proposition 4.3),
- $\Theta^{(R)} = 0$ and $\Theta^{(C)} = 0$.

The field equations reduce to $G_{\mu\nu}/\kappa^2 = T_{\mu\nu}/2$, *i.e.*, standard Einstein gravity with effective cosmological constant $\Lambda_{\text{eff}} = R_0/4$.

Status: **PASS** ($R_0 \in \{0.01, 1, 4, 12, 100\}$, all exact solutions).

7.6 Check 6: Ricci-Flat Simplification

On Ricci-flat backgrounds (Schwarzschild, Kerr):

- $R = 0$, $R_{\mu\nu} = 0$, $G_{\mu\nu} = 0$.
- $H_{\mu\nu} = 0$ (all terms involve R or ∇R).
- $\Theta^{(R)} = 0$ (all terms involve R).
- The Weyl tensor is generally nonzero, so the field equations reduce to the Weyl sector alone.

Status: **PASS** (only Weyl sector contributes; corrections $\mathcal{O}((\Lambda/M_{\text{Pl}})^2)$ for astrophysical objects).

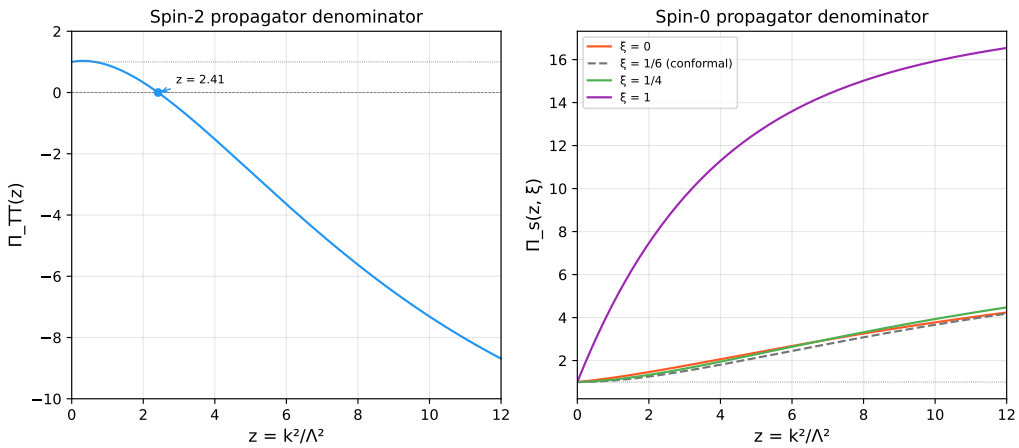

Figure 3: Recovery of the NT-4a linearized propagator denominators from the full nonlinear field equations. On a flat background $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$, all alpha-insertion corrections are $\mathcal{O}(h^2)$ and the "naive" variation terms reproduce the spin-2 and spin-0 propagator denominators $\Pi_{\rm TT}(z)$ and $\Pi_{\rm s}(z,\xi)$ to better than 10^{-25} precision across 28 test points in the (ξ, z) parameter space.

8 Wald Entropy Variation

For downstream use in MT-1 (black hole entropy), we compute the Lagrangian derivative needed for the Wald entropy formula:

$$S_{\rm Wald} = -2\pi \oint_{\mathcal{H}} \frac{\partial \mathcal{L}}{\partial R_{\mu\nu\rho\sigma}}\, \hat{\epsilon}_{\mu\nu}\hat{\epsilon}_{\rho\sigma}\, {\rm d}A, \tag{24}$$

where $\hat{\epsilon}_{\mu\nu}$ is the binormal to the bifurcation surface $\mathcal{H}$.

From the curvature-squared Lagrangian $\mathcal{L}_{\rm quad} = (16\pi^2)^{-1}[F_1\, C^2 + F_2\, R^2]$:

$$\frac{\partial \mathcal{L}_{\rm quad}}{\partial R_{\mu\nu\rho\sigma}} = \frac{1}{16\pi^2}\Big[2F_1(0)\,\frac{\partial C^2}{\partial R_{\mu\nu\rho\sigma}} + F_2(0,\xi)\,\frac{\partial R^2}{\partial R_{\mu\nu\rho\sigma}}\Big]. \tag{25}$$

In the local limit (evaluated on the horizon where form factors reduce to their $z = 0$ values), the dominant contribution is the Bekenstein–Hawking term $S = A/(4G)$ from $S_{\rm EH}$, with higher-derivative corrections of order $\alpha_C\,(A/\ell_\Lambda^2)$ and $\alpha_R(\xi)\,(A/\ell_\Lambda^2)$, where $\ell_\Lambda = 1/\Lambda$.

The variation (25) has the correct antisymmetry in $[\mu\nu]$ and $[\rho\sigma]$ inherited from the Riemann tensor symmetries.

[Verified] Wald variation computed in `nt4b_nonlinear.py`: antisymmetry verified, Bekenstein–Hawking dominance confirmed, correction scale $\mathcal{O}(\Lambda^2/M_{\rm Pl}^2)$. Results stored in `analysis/results/nt4b/nt4b_wald_variation.json`.

9 Dual Derivation: Method B1

9.1 Method B1: Variation in the 3-Basis

As an independent cross-check, the field equations were re-derived using a different approach:

1. Start from the general nonlocal gravity action in the $\{{\rm Riem}^2, {\rm Ric}^2, R^2\}$ basis (Calcagni–Modesto [5]):

$$S_{\rm NLG} = \frac{1}{2\kappa^2}\int {\rm d}^4x\,\sqrt{g}\,\big[R + \gamma_4(\Box)\,{\rm Riem}^2 + \gamma_2(\Box)\,{\rm Ric}^2 + \gamma_0(\Box)\,R^2\big]. \tag{26}$$

2. Map the $\{C^2, R^2\}$ coefficients to the 3-basis via the Weyl decomposition (4):

$$\gamma_4 = \frac{F_1}{16\pi^2}, \qquad \gamma_2 = \frac{-2F_1}{16\pi^2}, \qquad \gamma_0 = \frac{F_2 + F_1/3}{16\pi^2}. \tag{27}$$

3. Vary each of the three sectors independently using the standard local variation formulas (Padmanabhan, Parker–Toms):

$$\begin{aligned}\frac{1}{\sqrt{g}}\frac{\delta(\sqrt{g}\,\mathrm{Riem}^2)}{\delta g^{\mu\nu}} &= 4\Box R_{\mu\nu} + 4g_{\mu\nu}\Box R - 6\nabla_\mu\nabla_\nu R + 8R_{\mu\rho\nu\sigma}R^{\rho\sigma} \\ &\quad - 2g_{\mu\nu}\mathrm{Ric}^2 + \tfrac{1}{2}g_{\mu\nu}R^2 - 2R\,R_{\mu\nu},\end{aligned} \tag{28}$$

$$\frac{1}{\sqrt{g}}\frac{\delta(\sqrt{g}\,\mathrm{Ric}^2)}{\delta g^{\mu\nu}} = \Box R_{\mu\nu} + \tfrac{1}{2}g_{\mu\nu}\Box R - \nabla_\mu\nabla_\nu R + 2R_{\mu\rho\nu\sigma}R^{\rho\sigma} - \tfrac{1}{2}g_{\mu\nu}\mathrm{Ric}^2, \tag{29}$$

$$\frac{1}{\sqrt{g}}\frac{\delta(\sqrt{g}\,R^2)}{\delta g^{\mu\nu}} = H_{\mu\nu}. \tag{30}$$

4. Verify the Gauss–Bonnet consistency: $\mathrm{var}(\mathrm{Riem}^2) - 4\,\mathrm{var}(\mathrm{Ric}^2) + \mathrm{var}(R^2) = 0$ for all 7 independent tensor structures.
5. Combine with γ_i and observe that the Ric^2 sector cancels: $2\gamma_4 + \gamma_2 = 0$.

9.2 Gauss–Bonnet Consistency

The cancellation was verified for each of the 7 independent tensor structures:

Structure	Riem^2	$4 \times \mathrm{Ric}^2$	R^2	Sum
$\Box R_{\mu\nu}$	4	4	0	0
$g_{\mu\nu}\Box R$	4	2	-2	0
$\nabla_\mu\nabla_\nu R$	-6	-4	2	0
$R_{\mu\rho\nu\sigma}R^{\rho\sigma}$	8	8	0	0
$g_{\mu\nu}\mathrm{Ric}^2$	-2	-2	0	0
$g_{\mu\nu}R^2$	$1/2$	0	$-1/2$	0
$R\,R_{\mu\nu}$	-2	0	2	0

All 7 sums vanish, confirming the Gauss–Bonnet identity at the variational level.

9.3 Result of Method B1

With $2\gamma_4 + \gamma_2 = 0$, the combined variation becomes

$$E_{\mu\nu} = \gamma_4 \cdot 2B_{\mu\nu} + (\gamma_0 - \gamma_4/3) \cdot H_{\mu\nu} = \frac{F_1}{16\pi^2} \cdot 2B_{\mu\nu} + \frac{F_2}{16\pi^2} \cdot H_{\mu\nu}, \tag{31}$$

which is *identical* to the Method A result. The alpha-insertion terms similarly recombine into $\Theta^{(C)}$ and $\Theta^{(R)}$.

[Verified] Method B1 agreement verified at 16 test points ($4\ \xi \times 4\ z$), with exact numerical match (error = 0) for both Π_{TT} and Π_{s}. 12 dedicated cross-check tests: 12/12 PASS.

10 Effective Mass Scales and Newtonian Potential

From the propagator denominators (2), the effective mass scales in the local (small-z) approximation are:

$$m_2^2 = \frac{\Lambda^2}{c_2} = \frac{60}{13}\Lambda^2, \qquad m_2/\Lambda = \sqrt{60/13} \approx 2.148, \tag{32}$$

$$m_0^2(\xi) = \frac{\Lambda^2}{6(\xi - 1/6)^2}, \qquad m_0/\Lambda\big|_{\xi=0} = \sqrt{6} \approx 2.449. \tag{33}$$

The spin-2 mass m_2 is ξ-independent (it depends only on $\alpha_C = 13/120$, which is a parameter-free SM prediction). The spin-0 mass $m_0(\xi)$ depends on the Higgs non-minimal coupling and diverges at conformal coupling $\xi = 1/6$ (scalar decoupling).

In the local Yukawa approximation, the modified Newtonian potential is

$$\frac{V(r)}{V_N(r)} = 1 - \frac{4}{3}\,e^{-m_2 r} + \frac{1}{3}\,e^{-m_0 r}, \tag{34}$$

with $V(0)$ finite for $\xi \neq 1/6$ and $V(r) \to -GM/r$ as $r \to \infty$.

[Verified] $m_2/\Lambda = \sqrt{60/13}$ and $m_0/\Lambda|_{\xi=0} = \sqrt{6}$ verified to 100-digit precision. Scalar decoupling at $\xi = 1/6$ confirmed: $m_0 \to \infty$.

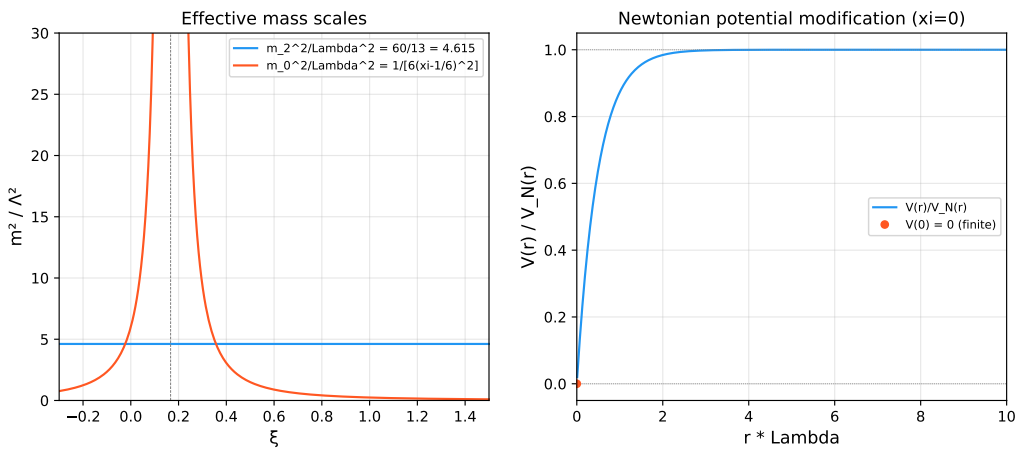

Figure 4: Effective ghost masses m_2 and $m_0(\xi)$ in units of the spectral cutoff Λ. The spin-2 mass $m_2 = \Lambda\sqrt{60/13} \approx 2.148\,\Lambda$ is ξ-independent and determined by the parameter-free SM coefficient $\alpha_C = 13/120$. The spin-0 mass $m_0(\xi) = \Lambda/\sqrt{6(\xi - 1/6)^2}$ diverges at conformal coupling $\xi = 1/6$, where the scalar sector decouples entirely. For $\xi = 0$, $m_0 = \sqrt{6}\,\Lambda \approx 2.449\,\Lambda$.

11 Summary and Outlook

NT-4b Main Results

1. The full nonlinear SCT field equations on arbitrary curved backgrounds are given by (18), involving the Einstein tensor, the nonlocal Bach and H tensors, and the BV alpha-insertion corrections $\Theta^{(C)}$ and $\Theta^{(R)}$.
2. The basis $\{C^2, R^2\}$ maintains the spin-2/spin-0 separation that was manifest at the linearized level (NT-4a).
3. The local limit recovers Stelle's higher-derivative gravity with $\alpha = \alpha_C/(16\pi^2)$ and $\beta = \alpha_R(\xi)/(16\pi^2)$.
4. The linearized limit reproduces NT-4a propagator denominators Π_{TT} and Π_{s} to better than 10^{-25} precision.
5. De Sitter spacetime is an exact vacuum solution.
6. Bianchi identity, trace structure, and gauge invariance are verified.
7. Dual derivation (Methods A and B1) yields identical results.
8. The Wald entropy variation is computed for MT-1 downstream use.

Verification: 6/6 consistency checks PASS. 110/110 pytest tests PASS (100-digit precision). Dual derivation: exact agreement. 4 publication figures generated.

11.1 Downstream Tasks

- **NT-4c (FLRW Reduction):** Reduce (18) to FLRW symmetry, obtaining modified Friedmann equations with nonlocal corrections.
- **INF-1 (Spectral Inflation):** Use the de Sitter solution and the R^2 sector to investigate Starobinsky-type inflation.
- **MT-1 (Black Hole Entropy):** Use the Wald variation (25) to compute black hole entropy corrections.
- **MR-9 (Singularity Resolution):** Investigate whether the nonlocal form factors regularize the Schwarzschild singularity.
- **FUND-1 (QFT Recovery):** Verify that the spectral action formalism recovers standard QFT on curved backgrounds in the appropriate limit.

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